Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3391-WE01

Title: (BEng/MEng)

Control and Signal Processing 3

Time Allowed:	2 hours
Additional Material provided:	Yes, formula sheet included as part of exam paper
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Section A

Question 1

(a) A continuous-time signal $x(t)$ has frequency components at 5 kHz, 7 kHz and 10 kHz. The signal $x(t)$ is sampled with period *T* and its spectrum is computed using a fast Fourier transform (FFT) algorithm which requires the number of samples *N* to be a power of two. Determine the minimum value of the sampling rate such that aliasing is avoided.

[10%]

(b) A system operates on two signals $x[n]$ and $y[n]$ to give the signal $z[k]$ as in Eqn. 1.

$$z[k] = \sum_{n=0}^{N-1} x[n]y[n+k] \qquad \text{(Eqn. 1)}$$

Find $z[k]$ for $x[n]$ = {1 3 5 6} and $y[n]$ = {2 6 7}.

[20%]

(c) For the system shown in Figure Q.1, write an expression for the output $y(t)$ in terms of the input $x(t)$ and find the impulse response, $h(t)$. Determine whether the system is causal.

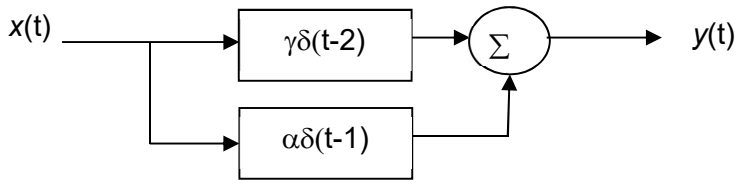


Figure Q.1

[40%]

(d) A filter has an impulse response given by Eqn. 2.

$$h(t) = 1 \qquad 0 < t < T \qquad \text{(Eqn. 2)}$$

(i) Derive the frequency response of the filter.
(ii) Sketch the magnitude and phase of its frequency response.

[30%]

Question 2

(a) Given the signal $x(t)$ defined by Eqn. 3,

$$x(t) = cos\omega_o t + sin^2 2\omega_o t + 3sin\omega_o t.cos3\omega_o t + 8 \quad \text{(Eqn. 3)}$$

(i) Express $x(t)$ in terms of a complex exponential Fourier series.

[20%]

(ii) Express it in terms of the compact trigonometric series identifying the magnitude and the phase.

[15%]

(b) Find an analytical expression for the convolution of $x(t)$ and $y(t)$ given by Eqn. 4.

$$x(t) = e^{-t}u(t-2)$$
$$y(t) = u(t-4) \quad \text{(Eqn. 4)}$$

[25%]

(c) The nonlinear system defined by Eqn. 5

$$y(t) = x(t) + 0.4\,x^2(t) + 4 \quad \text{(Eqn. 5)}$$

has an input signal with the bandpass spectrum given in Eqn. 6

$$X(\omega) = 4\,\frac{rect(\omega-20)}{10} + 4\,\frac{rect(\omega+20)}{10} \quad \text{(Eqn.6)}$$

where

$$\frac{rect(\omega)}{B} = 1 \qquad -\frac{B}{2} < \omega < \frac{B}{2}$$

(i) Sketch the spectrum of the input signal $X(\omega)$, labelling all important frequencies and amplitudes.
(ii) Give an expression for $Y(\omega)$ identifying each output.
(iii) Sketch the frequency range of the output of the system.

[40%]

continued

Section B

Question 3

A plant process is described by the following transfer function,

$$G(s) = \frac{3}{(s+1)(s+4)}$$

together with a negative unit feedback link, they form a closed-loop system as shown in Figure Q3.1.

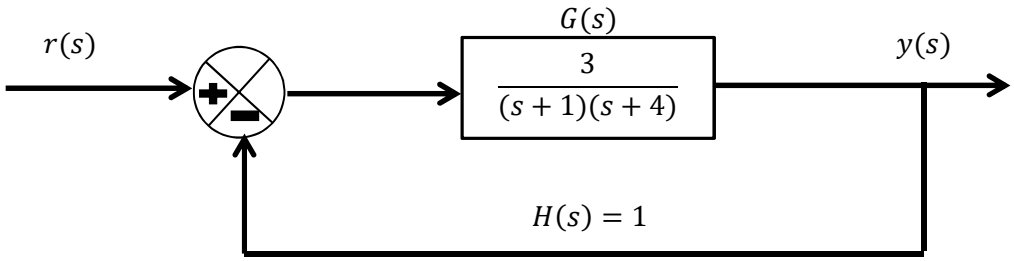


Figure Q.3.1: A closed-loop system with negative unit feedback

(a) Design a practical phase-lead compensator $D(s) = k\frac{s+z}{s+p}$ in series with $G(s)$ to satisfy the performance specifications of damping ratio $\zeta = 0.8$ and the settling time of 2% settling error to be less than 1 second.

[30%]

(b) Analyse in your own words the features of two controllers: the designed compensator in (a), and a PD controller with the transfer function of $10.67 + 1.53s$.

[30%]

(c) Discretise this designed compensator in (a) by using the backward difference algorithm, i.e., $z = \frac{1}{1-sT}$ (assuming the sampling interval *T=0.2* second) that is suitable for being implemented in a microprocessor.

[20%]

(d) Discuss in your own words the impacts of choosing different sampling intervals when discretising the designed compensator in (a), for example, considering the sampling interval *T=0.2* second and that of *T=1* second.

[20%]

Question 4

A plant process is described by the following transfer function:

$$G(s) = \frac{1}{s(s+1)(4s+1)}$$

and is placed in a closed-loop control system as shown in Figure Q4.1.

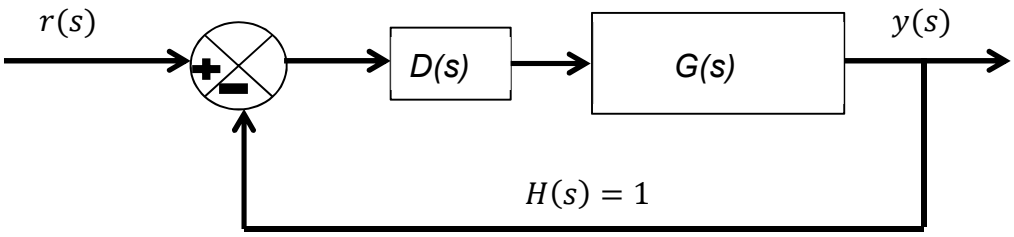


Figure Q4.1: A closed-loop control system with negative unit feedback

(a) If the controller *D(s)* in Figure Q4.1 is configured as a gain controller,

(i) Use the provided graph paper to sketch the root locus of this closed-loop system with necessary steps of the root locus plotting.

[20%]

(ii) Find the critical value of the gain that makes the system become marginally stable.

[20%]

(iii) If the controller *D(s)=1,* write the open-loop frequency response formula $GH(i\omega)$ in the complex number form of $a + ib$. Calculate its open-loop frequency response as the frequency ω varies, and present your data using a Nyquist diagram.

[30%]

(b) If the compensator *D(s)* in Figure Q4.1 can be configured as ideal phase-lead compensators where an ideal phase-lead compensator has the form of *D(s)= k(s+z)*, the system can be more stable. To satisfy a performance specification of damping ratio $\zeta = 0.5$ and natural frequency $\omega_n = 4$ rad/s, use both angle and magnitude criteria to design the compensators. Discuss the challenging issues during this design.

[30%]

L3 Signal Processing Formula Sheet

Energy of a signal

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

Power of a signal

$$P_{av} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

Fourier series

$$x(t) = a_o + \sum_{k=1}^{\infty} \left[a_k \cos k\omega_o t + b_k \sin k\omega_o t \right]$$

where

$$a_o = \frac{1}{T} \int_T x(t)dt = dc\ term$$

$$a_k = \frac{2}{T} \int_T x(t) \cos k\omega_o t dt \qquad b_k = \frac{2}{T} \int_T x(t) \sin k\omega_o t dt$$

Alternative Fourier series

Trigonometric form:

$$x(t) = a_o + \sum_{k=1}^{\infty} \sqrt{a_k^2 + b_k^2} \cos(k\omega_o t + \phi_k)$$

where

$$\phi_k = -\tan^{-1} \frac{b_k}{a_k}$$

Complex exponential form

$$x(t) = \sum_{k=-\infty}^{\infty} d_k\, e^{jk\omega_o t}$$

where $d_k = \frac{1}{T} \int_T x(t).e^{-jk\omega_o t} dt$

and $d_k = \frac{1}{2}(a_k - jb_k)\ for\ k > 0\ and\ d_k = \frac{1}{2}(a_k + jb_k)\ for\ k < 0$

continued

Fourier transform

$$\mathcal{F}\{x(t)\} = X(\omega)$$
$$= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

Alternatively, it can be written in terms of the frequency in cycles per seconds or Hz as

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$

Inverse Fourier transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

or

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

Fourier transform properties

1. Linearity

$$x_1(t) \leftrightarrow X_1(\omega), \text{and } x_2(t) \leftrightarrow X_2(\omega) \text{ then}$$
$$ax_1(t) + bx_2(t) \leftrightarrow aX_1(\omega) + bX_2(\omega)$$

2. Differentiation

$$y(t) = \frac{d^n x(t)}{dt^n} \leftrightarrow (j\omega)^n X(\omega)$$

3. Integration

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau \leftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

4. Time scaling

$$x(bt) \leftrightarrow \frac{1}{|b|}X\left(\frac{\omega}{b}\right)$$

5. Time reversal

$$x(-t) \leftrightarrow X(-\omega) = X^*(\omega)$$

6. Time delay

$$y(t) = x(t-T) \leftrightarrow X(\omega)\, e^{-j\omega T}$$

7. Multiplication in time

continued

$$x_1(\mathrm{t}).x_2(\mathrm{t}) \leftrightarrow \frac{X_1(\omega) * X_2(\omega)}{2\pi}$$

8. Multiplication in frequency (convolution in time)

$$X_1(\omega)X_2(\omega) \leftrightarrow x_1(\mathrm{t}) * x_2(\mathrm{t})$$

9. Modulation property

$$x(\mathrm{t})\cos(\omega_o t) \leftrightarrow \frac{1}{2}X(\omega - \omega_o) + \frac{1}{2}X(\omega + \omega_o)$$

10. Duality property

If $x(t) \leftrightarrow X(\omega)$

Then $X(t) \leftrightarrow 2\pi x(-\omega)$

Discrete Fourier Transform

Discrete Fourier Series

$$d_k = X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{\frac{-j2\pi k}{N}}$$

Inverse Discrete Fourier Series

IDFS: $$x[n] = \sum_{k=0}^{N-1} d_k\, e^{\frac{j2\pi kn}{N}}$$

Euler's identities

$$A\sin(\omega_o t) = \frac{A}{j2}\left(e^{j\omega_o t} - e^{-j\omega_o t}\right)$$

$$A\cos(\omega_o t) = \frac{A}{2}\left(e^{j\omega_o t} + e^{-j\omega_o t}\right)$$

$$A\cos(\omega_o t + \phi_o) = \frac{A}{2}\left(e^{j\omega_o t}e^{j\phi_o} + e^{-j\omega_o t}e^{-j\phi_o}\right)$$

$e^{\pm j\beta} = \cos\beta \pm j\sin\beta$

Impulse function in continuous time domain

$\delta(t) = 0\ for\ t \neq 0\ and\ infinite\ at\ t = 0$

$$\int_{t=-\infty}^{\infty} \delta(t) = 1$$

Impulse function in discrete time domain

continued

$\delta[n] = 1\ n = 0\ and\ 0\ elsewhere$

Unit step function in continuous time domain

$u(t) = 1\ for\ t \geq 0\ and\ 0\ t < 0$

Unit step function in discrete time domain

$u[n] = 1\ for\ n \geq 0\ and\ 0\ for\ n < 0$

Convolution integral

$$y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau = \int_{-\infty}^{\infty} x_1(t-\tau)x_2(\tau)d\tau$$

Convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] = \sum_{k=-\infty}^{\infty} x_1[n-k]x_2[k]$$

Convolution with an impulse

$x(t) * \delta(t - t_o) = x(t - t_o)$

Correlation functions

Fourier transform of an impulse

$\delta(t) \leftrightarrow 1$

Inverse Fourier transform of an impulse

$\delta(\omega) \leftrightarrow 1/2\pi$

Roots of a second order polynomial

$ax^2 + bx + c = 0$

are given by

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometric identities

cos θ = cos(-θ) = sin(π/2 - θ)

sin θ = - sin(-θ) = cos(π/2 - θ)

tan θ = - tan(-θ) = cot(π/2 - θ)

sin 2θ = 2 cos θ sin θ

cos 2θ = $\cos^2\theta$ - $\sin^2\theta$ = 2 $\cos^2$ θ- 1 = 1 - 2 $\sin^2\theta$

continued

tan 2θ = 2 tan θ/ (1 - tan2θ)

$\sin\alpha.\sin\beta = \frac{1}{2}\{(\cos(\alpha-\beta) - cos(\alpha+\beta)\}$

$\cos\alpha.\cos\beta = \frac{1}{2}\{(\cos(\alpha-\beta) + cos(\alpha+\beta)\}$

$\sin\alpha.\cos\beta = \frac{1}{2}\{(\sin(\alpha-\beta) + sin(\alpha+\beta)\}$

sin θ/2 = ± √[(1 - cos θ)/2]

cos θ/2 = ± √[(1 + cos θ)/2]

tan θ/2 = sin θ/(1 + cos θ)

$\cos^2\theta + \sin^2\theta = 1$

$\sec^2\theta - \tan^2\theta = 1$

$\text{cosec}^2\theta - \cot^2\theta = 1$

$\sin^2\theta = ½(1 - \cos 2\theta)$

$\cos^2\theta = ½(1 + \cos 2\theta)$

$\tan^2\theta = (1 - \cos 2\theta)/(1 + \cos 2\theta)$

sin(A + B) = sin A cos B + cos A sin B

sin(A - B) = sin A cos B - cos A sin B

cos(A + B) = cos A cos B - sin A sin B

cos(A - B) = cos A cos B + sin A sin B

tan(A + B) = (tan A + tan B)/(1 - tan A tan B)

tan(A - B) = (tan A - tan B)/(1 + tan A tan B)

sin A + sin B = 2 sin ½(A + B) cos ½(A - B)

sin A - sin B = 2 cos ½(A + B) sin ½(A - B)

cos A + cos B = 2 cos ½(A + B) cos ½(A - B)

cos A - cos B = - 2 sin ½(A + B) sin ½(A - B)

tan A + tan B = sin (A + B)/(cos A cos B)

tan A - tan B = sin (A - B)/(cos A cos B)

sin2A + sin2B = 1 - cos(A + B)cos(A - B)

sin2A - sin2B = sin(A + B)sin(A - B)

continued

cos2A + sin2B = 1 - sin(A + B)sin(A - B)

cos2A - sin2B = cos(A + B)cos(A - B)

cos2A + cos2B = 1 + cos(A + B)cos(A - B)

cos2A - cos2B = - sin(A + B)sin(A - B)

For a triangle with sides a, b, c, and angles A, B, C opposite sides a, b and c respectively, the following relations hold.

a2 = b2 + c2 - 2 b c cos A

a/sin A = b/sin B = c/sin C.

(a - b)/(a + b) = tan ½(A - B)/ tan ½(A + B)

L3 Control Formula Sheet

Standard second-order transfer function

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Closed-loop poles of under-damped second-order system

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The overshoot percentage (%) of the standard second-order system

$$M_o = 100\exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

The settling time for a second-order system can be defined as

$$T_s = \frac{\text{Number of time constants}}{\zeta\omega_n}$$

The relation of number of time constants and settling error

Number of time constants	$e^{-\frac{t}{\tau}}$	Settling Error (%)
1	0.368	36.8
2	0.135	13.5
3	0.05	5
4	0.018	1.8
5	0.007	0.7

Defining $u(t)$ as the controller output and $e(t)$ as the error, the standard form of the PID algorithm is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{d}{dt}e(t)$$

Laplace form of the PID controller

$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

7 Steps for Root Locus Sketching:

1). Write the characteristic function so that *K* is a multiplier *1+KG(s)=0.*

2). Factor *G(s)* in terms of *n* poles and *m* zeros:

continued

$$G(s) = \frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{n}(s - p_j)}$$

3). Mark poles and zeros on the *s*-plane.

4). Determine the number of separate loci (lines)

5). Locate the segments of the real axis that are root loci

6). The remaining loci proceed to the zeros at infinity along asymptotes centered at σ and with angle Φ

$$\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n - m}$$

$$\phi = \frac{(2q+1)}{n-m}\pi,$$

where $q = 0, 1, \ldots, (n - m - 1)$

7). Plot remaining root locus.

Magnitude Criteria:

$$K = \frac{\prod_{i=1}^{n}|s - p_i|}{\prod_{j=1}^{m}|s - z_j|}$$

Angle Criteria:

$$\sum_{i=1}^{n}\phi_{p_i} - \sum_{j=1}^{n}\phi_{z_i} = (2q+1)\pi$$

END